Zebra puzzle

Zebra puzzle is a logic puzzle in which the solution is a set of five houses, each with a different color, inhabited by people of different nationality, with different pets, drinking different drinks, and smoking different brands of cigarettes.

Color	Nationality	Pet	Drink	Cigarette

Rules

- The Norwegian lives in the first house.
- The Ukrainian drinks tea.
- The Lucky Strike smoker drinks orange juice.
- The Japanese smokes Parliaments.
- The Spanish keeps a dog.
- The Kent smoker lives in the red house.
- The owner of the green house drinks coffee.
- The Chesterfield smoker lives next to the fox owner.
- The owner of the green house is immediately to the right of the ivory house.
- The Old Gold smoker keeps snails.
- The Norwegian lives next to the blue house.
- The Kent smoker has a neighbor who drinks water.
- The owner of the yellow house smokes Kools.
- The owner who keeps the horse lives next to the owner who drinks tea.
- The owner who keeps the horse is the Ukrainian.
- The Lucky Strike smoker drinks orange juice.
- The Japanese smokes Parliaments.
- The Spanish keeps a dog.
- The Norwegian lives next to the blue house.

GPT-Speed Instructions

GPT-Speed Instructions are a set of instructions for the GPT-Speed machine, a hypothetical machine that can execute GPT-Speed programs. GPT-Speed Instructions are a set of instructions that can be executed by the GPT-Speed machine. GPT-Speed Instructions are a set of instructions that can be executed by the GPT-Speed machine.

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    - 19.2.18 Spanish keeps a dog
    - 19.2.19 Norwegian lives next to blue house
- 20. Conversion of the task as a constraint problem
  - 20.1 Variables
    - 20.1.1 Color variables
    - 20.1.2 Nationality variables
    - 20.1.3 Pet variables
    - 20.1.4 Drink variables
    - 20.1.5 Cigarette variables
    - 20.1.6 Position variables
    - 20.1.7 Auxiliary variables
  - 20.2 Constraints
    - 20.2.1 Norwegian lives next to blue house
    - 20.2.2 Ukrainian drinks tea
    - 20.2.3 Lucky Strike smoker drinks orange juice
    - 20.2.4 Japanese smokes Parliaments
    - 20.2.5 Spanish keeps a dog
    - 20.2.6 Kent smoker lives in red house
    - 20.2.7 Green house drinks coffee
    - 20.2.8 Chesterfield smoker lives next to fox owner
    - 20.2.9 Green house is immediately to the right of ivory house
    - 20.2.10 Old Gold smoker keeps snails
    - 20.2.11 Norwegian lives next to blue house
    - 20.2.12 Kent smoker has a neighbor who drinks water
    - 20.2.13 Yellow house smokes Kools
    - 20.2.14 Owner who keeps horse lives next to owner who drinks tea
    - 20.2.15 Owner who keeps horse is the Ukrainian
    - 20.2.16 Lucky Strike smoker drinks orange juice
    - 20.2.17 Japanese smokes Parliaments
    - 20.2.18 Spanish keeps a dog
    - 20.2.19 Norwegian lives next to blue house
- 21. Conversion of the task as a constraint problem
  - 21.1 Variables
    - 21.1.1 Color variables
    - 21.1.2 Nationality variables
    - 21.1.3 Pet variables
    - 21.1.4 Drink variables
    - 21.1.5 Cigarette variables
    - 21.1.6 Position variables
    - 21.1.7 Auxiliary variables
  - 21.2 Constraints
    - 21.2.1 Norwegian lives next to blue house
    - 21.2.2 Ukrainian drinks tea
    - 21.2.3 Lucky Strike smoker drinks orange juice
    - 21.2.4 Japanese smokes Parliaments
    - 21.2.5 Spanish keeps a dog
    - 21.2.6 Kent smoker lives in red house
    - 21.2.7 Green house drinks coffee
    - 21.2.8 Chesterfield smoker lives next to fox owner
    - 21.2.9 Green house is immediately to the right of ivory house
    - 21.2.10 Old Gold smoker keeps snails
    - 21.2.11 Norwegian lives next to blue house
    - 21.2.12 Kent smoker has a neighbor who drinks water
    - 21.2.13 Yellow house smokes Kools
    - 21.2.14 Owner who keeps horse lives next to owner who drinks tea
    - 21.2.15 Owner who keeps horse is the Ukrainian
    - 21.2.16 Lucky Strike smoker drinks orange juice
    - 21.2.17 Japanese smokes Parliaments
    - 21.2.18 Spanish keeps a dog
    - 21.2.19 Norwegian lives next to blue house
- 22. Conversion of the task as a constraint problem
  - 22.1 Variables
    - 22.1.1 Color variables
    - 22.1.2 Nationality variables
    - 22.1.3 Pet variables
    - 22.1.4 Drink variables
    - 22.1.5 Cigarette variables
    - 22.1.6 Position variables
    - 22.1.7 Auxiliary variables
  - 22.2 Constraints
    - 22.2.1 Norwegian lives next to blue house
    - 22.2.2 Ukrainian drinks tea
    - 22.2.3 Lucky Strike smoker drinks orange juice
    - 22.2.4 Japanese smokes Parliaments
    - 22.2.5 Spanish keeps a dog
    - 22.2.6 Kent smoker lives in red house
    - 22.2.7 Green house drinks coffee
    - 22.2.8 Chesterfield smoker lives next to fox owner
    - 22.2.9 Green house is immediately to the right of ivory house
    - 22.2.10 Old Gold smoker keeps snails
    - 22.2.11 Norwegian lives next to blue house
    - 22.2.12 Kent smoker has a neighbor who drinks water
    - 22.2.13 Yellow house smokes Kools
    - 22.2.14 Owner who keeps horse lives next to owner who drinks tea
    - 22.2.15 Owner who keeps horse is the Ukrainian
    - 22.2.16 Lucky Strike smoker drinks orange juice
    - 22.2.17 Japanese smokes Parliaments
    - 22.2.18 Spanish keeps a dog
    - 22.2.19 Norwegian lives next to blue house
- 23. Conversion of the task as a constraint problem
  - 23.1 Variables
    - 23.1.1 Color variables
    - 23.1.2 Nationality variables
    - 23.1.3 Pet variables
    - 23.1.4 Drink variables
    - 23.1.5 Cigarette variables
    - 23.1.6 Position variables
    - 23.1.7 Auxiliary variables
  - 23.2 Constraints
    - 23.2.1 Norwegian lives next to blue house
    - 23.2.2 Ukrainian drinks tea
    - 23.2.3 Lucky Strike smoker drinks orange juice
    - 23.2.4 Japanese smokes Parliaments
    - 23.2.5 Spanish keeps a dog
    - 23.2.6 Kent smoker lives in red house
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    - 23.2.10 Old Gold smoker keeps snails
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    - 23.2.12 Kent smoker has a neighbor who drinks water
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    - 23.2.14 Owner who keeps horse lives next to owner who drinks tea
    - 23.2.15 Owner who keeps horse is the Ukrainian
    - 23.2.16 Lucky Strike smoker drinks orange juice
    - 23.2.17 Japanese smokes Parliaments
    - 23.2.18 Spanish keeps a dog
    - 23.2.19 Norwegian lives next to blue house
- 24. Conversion of the task as a constraint problem
  - 24.1 Variables
    - 24.1.1 Color variables
    - 24.1.2 Nationality variables
    - 24.1.3 Pet variables
    - 24.1.4 Drink variables
    - 24.1.5 Cigarette variables
    - 24.1.6 Position variables
    - 24.1.7 Auxiliary variables
  - 24.2 Constraints
    - 24.2.1 Norwegian lives next to blue house
    - 24.2.2 Ukrainian drinks tea
    - 24.2.3 Lucky Strike smoker drinks orange juice
    - 24.2.4 Japanese smokes Parliaments
    - 24.2.5 Spanish keeps a dog
    - 24.2.6 Kent smoker lives in red house
    - 24.2.7 Green house drinks coffee
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    - 24.2.9 Green house is immediately to the right of ivory house
    - 24.2.10 Old Gold smoker keeps snails
    - 24.2.11 Norwegian lives next to blue house
    - 24.2.12 Kent smoker has a neighbor who drinks water
    - 24.2.13 Yellow house smokes Kools
    - 24.2.14 Owner who keeps horse lives next to owner who drinks tea
    - 24.2.15 Owner who keeps horse is the Ukrainian
    - 24.2.16 Lucky Strike smoker drinks orange juice
    - 24.2.17 Japanese smokes Parliaments
    - 24.2.18 Spanish keeps a dog
    - 24.2.19 Norwegian lives next to blue house
- 25. Conversion of the task as a constraint problem
  - 25.1 Variables
    - 25.1.1 Color variables
    - 25.1.2 Nationality variables
    - 25.1.3 Pet variables
    - 25.1.4 Drink variables
    - 25.1.5 Cigarette variables
    - 25.1.6 Position variables
    - 25.1.7 Auxiliary variables
  - 25.2 Constraints
    - 25.2.1 Norwegian lives next to blue house
    - 25.2.2 Ukrainian drinks tea
    - 25.2.3 Lucky Strike smoker drinks orange juice
    - 25.2.4 Japanese smokes Parliaments
    - 25.2.5 Spanish keeps a dog
    - 25.2.6 Kent smoker lives in red house
    - 25.2.7 Green house drinks coffee
    - 25.2.8 Chesterfield smoker lives next to fox owner
    - 25.2.9 Green house is immediately to the right of ivory house
    - 25.2.10 Old Gold smoker keeps snails
    - 25.2.11 Norwegian lives next to blue house
    - 25.2.12 Kent smoker has a neighbor who drinks water
    - 25.2.13 Yellow house smokes Kools
    - 25.2.14 Owner who keeps horse lives next to owner who drinks tea
    - 25.2.15 Owner who keeps horse is the Ukrainian
    - 25.2.16 Lucky Strike smoker drinks orange juice
    - 25.2.17 Japanese smokes Parliaments
    - 25.2.18 Spanish keeps a dog
    - 25.2.19 Norwegian lives next to blue house
- 26. Conversion of the task as a constraint problem
  - 26.1 Variables
    - 26.1.1 Color variables
    - 26.1.2 Nationality variables
    - 26.1.3 Pet variables
    - 26.1.4 Drink variables
    - 26.1.5 Cigarette variables
    - 26.1.6 Position variables
    - 26.1.7 Auxiliary variables
  - 26.2 Constraints
    - 26.2.1 Norwegian lives next to blue house
    - 26.2.2 Ukrainian drinks tea
    - 26.2.3 Lucky Strike smoker drinks orange juice
    - 26.2.4 Japanese smokes Parliaments
    - 26.2.5 Spanish keeps a dog
    - 26.2.6 Kent smoker lives in red house
    - 26.2.7 Green house drinks coffee
    - 26.2.8 Chesterfield smoker lives next to fox owner
    - 26.2.9 Green house is immediately to the right of ivory house
    - 26.2.10 Old Gold smoker keeps snails
    - 26.2.11 Norwegian lives next to blue house
    - 26.2.12 Kent smoker has a neighbor who drinks water